

Hip and Knee Injuries

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Volker Schöffl in „Bamboozallad“, Thakekh, Laos, Photo Michael Simon

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11.1 Overview (Statistics)

Injuries and overuse damage to the hip, thigh, and knee joint have become more frequent in climbing and bouldering in recent years, with a significant increase of acute knee injuries [1, 2]. Changes within the sport and current trends in wall construction and route setting are considered to be the main cause for this change [1, 3]. Thereby, complex, three-dimensional wall structures and climbing routes often require unconventional physical effort of the athletes; gymnastic elements and new techniques are increasingly used. A comparative analysis of more than 600 climbing athletes recently showed an impressive increase of knee injuries between the survey periods of 1998–2001 and 2017–2018 [1, 2]. Thereby, acute knee injuries currently represent the fourth most common injury region in the sport, accounting for 7.1% of climbing-related injuries. Overuse injuries or other symptoms of the hip/groin (0.8%) or thigh (0.9%) are diagnosed less frequently [1] and are not among the ten most frequently injured body regions in climbing. However, a stringent diagnostic and therapeutic procedure is also of utmost importance for these complaints.

11.2 Diagnosis

The diagnostic procedure for complaints in the hip, thigh, or knee initially follows the usual criteria and should be structured as follows:

1. Anamnesis: How long have the complaints existed? After which strain or movement did the pain start? Can the complaints be triggered (e.g., specific body positions)? Do the complaints radiate? Have the pain characteristics changed? Which treatment has already been administered?
2. Clinical examination: Inspection of abnormalities (swelling, limping, etc.), palpation of the corresponding region (pain exacerbation by palpation, etc.), and joint-specific tests (e.g., FADIR test of the hip, meniscus tests,

etc.). This should always be accompanied by a comparative examination of the opposite side.

3. Imaging procedures: Ultrasound, X-ray, CT, and MRI, depending on the suspected diagnosis.

A detailed anamnesis and a detailed clinical examination are essential for the clarification of hip, thigh, and knee problems in climbers and usually allow a correct diagnosis. Imaging procedures can be selected based on the anamnesis and clinical examination. MRI is the preferable imaging technique for more precise clarification of the symptoms.

11.3 Femoroacetabular Impingement

Femoroacetabular impingement (FAI) is a common cause of hip pain in climbers and must be considered and recognized as it represents a major contributing factor to arthritis of the hip [4]. The underlying pathomechanism is usually a subtle deformation of the femoral neck (Fig. 11.1), whereby repeated abnormal bone contact (impingement) between the femoral neck and acetabulum occurs, particularly in cases of



Fig. 11.1 X-ray (lateral view) of a climbers right hip joint with radiological evidence of femoroacetabular impingement (red arrow marks the so-called “bump” at the femoral neck)

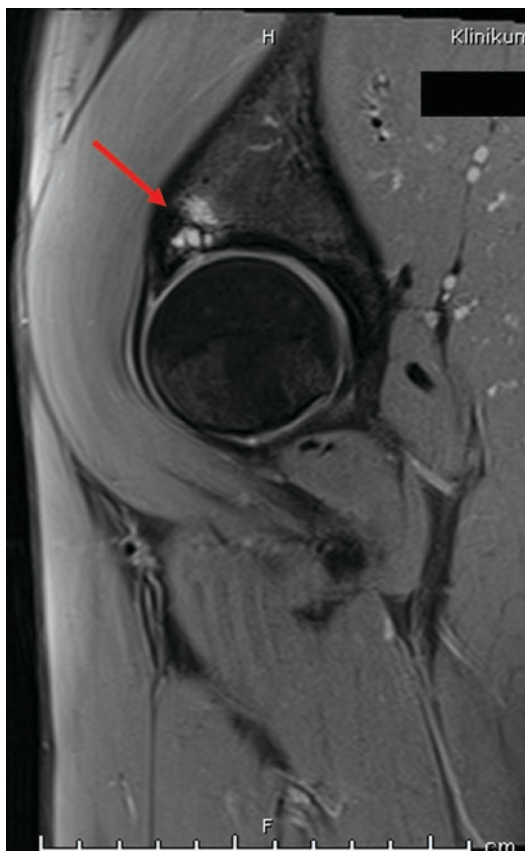


Fig. 11.2 MRI of a labral lesion with acetabular roof cysts (see arrow) in femoroacetabular impingement

strong flexion of the hip joint (e.g., use of the “high step” position), which can cause joint damage, pain, radiological changes, and arthritis [5] (Figs. 11.1 and 11.2). Two major types of FAI are classified. “Cam” impingement is a bony prominence (so-called “bump”) of the femoral head and neck transition (Fig. 11.1). During hip flexion and internal rotation, the bump is pressed against the acetabular rim, which can cause cartilage damage and labral tears (Fig. 11.2) [4, 5]. “Pincer” impingement is a protruding acetabulum. Flexion of the hip leads to contact between the femoral head and the acetabulum, which can also cause labral injuries and cartilage damage. Cam impingement is more common in younger, athletic men and is associated with reactive remodeling reactions of growth plates in children/

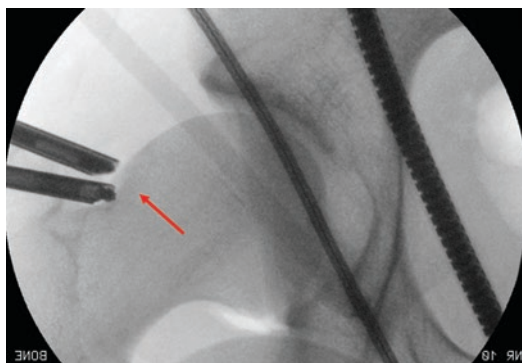


Fig. 11.3 Intraoperative fluoroscopy during arthroscopic femoral neck osteoplasty (see arrow) in femoroacetabular impingement

adolescents, who are very active in sports. Pincer impingement is usually more likely to be detected in middle-aged women. Mixed images are common [4].

Clinically, patients usually report pain during sports, especially when competing at high speed with maximum flexion and/or internal rotation of the hip (e.g., “drop knee” position). After a physical examination with appropriate stress tests (including the so-called FADIR test = flexion, adduction, and internal rotation of the hip), the radiological clarification (radiographs of the pelvis and the hip) and MR arthrography (Figs. 11.1 and 11.2) are performed. Depending on the findings, an arthroscopy of the affected hip joint should be considered (partial resection of the femoral bump, reconstruction of the labrum or arthroscopic cartilage therapy) (Fig. 11.3).

In the clarification of groin and hip pain, differential diagnoses must be considered both in the patient’s medical history and during the clinical examination. As shown in Fig. 11.4, acute injuries around the origin of the hip flexors/knee extensors can also trigger groin pain. The pictured athlete experienced severe pain in his groin while stepping on a high foot hold in an unfavorable body position. Magnetic resonance imaging confirmed a traumatic rectus femoris muscle tear with detachment of the pelvis (Fig. 11.4).

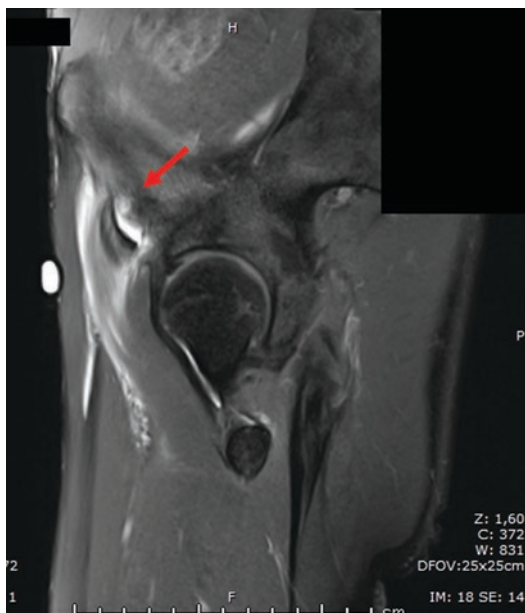


Fig. 11.4 MRI of an avulsion lesion of the rectus femoris muscle (see arrow)

- **Etiology:** Anatomical variants of the femoral head/neck transition and/or the acetabulum.
- **Onset:** Gradual, progressive, and load-dependent complaints, especially in cases of strong flexion/internal rotation of the hip joint.
- **Clinic:** Dull and deep hip/groin pain.
- **Diagnosis:** Clinical, radiographic, and MR arthrography.
- **Therapy:** Depending on the labrum and cartilage, arthroscopy of the hip joint with femoral neck shaping, labral reconstruction, cartilage therapy, acetabular roof plastic, etc.
- **Result:** Often asymptomatic after surgical correction. Subsequent progress of the joint destruction cannot always be safely avoided.



Fig. 11.5 Heel hook position of the right lower extremity during bouldering with external rotation and flexion in the hip and knee joint

11.4 Heel Hook Injuries

An injury mechanism that has become increasingly common in recent years is the heel hook-related injury, which is attributable almost exclusively to bouldering [6] (Fig. 11.5). Modern bouldering gyms or competition facilities offer complex wall structures, which require the use of the legs and feet in a much more complex motion sequence than formerly known and performed [7]. The increasing popularity of outdoor bouldering also contributes to the accumulation of heel hook-related injuries (Fig. 11.5).

To stabilize the body, especially in bouldering, heel or toe hooks are often used. In steep wall sections, the heels or toes are thereby used to transfer the tensile load of the ischiocrural muscles to the corresponding wall structures to stabilize the body position (Fig. 11.5). Additionally, these movements often require an external rotation of the lower leg with maximum external rotation of the knee. This position can lead to the maximum load on collateral ligaments, lateral meniscus, posterior cruciate ligament, popliteus tendon, posterior joint capsule, iliotibial tractus, and ischiocrural muscles. In the case of heel hook

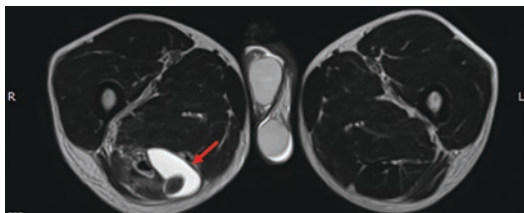


Fig. 11.6 MRI of an avulsion injury with hematoma formation in the ischiocrural muscles of the right lower extremity after acute trauma during heel hook (see arrow)

injuries, athletes often report a loud snapping sound in the event of the accident, which is most likely explained by a phenomenon in which the extended iliotibial tractus snaps over the lateral condyle, similar to a maximum pivot shift. The injuries can be divided into two main locations: injuries near the knee and near the pelvis. Injuries near the pelvis usually involve a tendon avulsion (partial or total) of the biceps femoris or the rectus femoris insertion (Fig. 11.6). Stretching techniques and sufficient warm-up might have a preventive effect. Clinical examination usually includes stress tests and palpation. In addition, sonography often provides indications of peritendinous or intramuscular hematomas around the ischiocrural muscles. Reliable visualization of tendon avulsions or muscle tears is also possible in some cases. To exclude any bony lesions, it is recommended to perform an X-ray diagnosis. For detailed evaluation of the muscle-tendon injuries, MRI is usually performed (Fig. 11.6). Depending on the extent of the muscle-tendon injury, conservative therapy is performed by resting and avoiding the corresponding load on the affected muscles. Higher graded lesions usually require a surgical procedure (Fig. 11.7). Ehiogu et al. recently evaluated current rehabilitation concepts for climbers after acute hamstring injuries [8]. The authors thereby highlighted the fact that rehabilitation should address the loading capacity of injured tissues early in the treatment regime using primarily concentric muscle-strengthening strategies.

Heel hook injuries close to the knee joint have already been described in other sports (e.g., martial arts), in which the opponent is often attacked



Fig. 11.7 Intraoperative findings after avulsion injury of the ischiocrural musculature. The figure shows the tendon stump sheared off from the pelvis before refixation

with the legs. A higher injury risk seems obvious due to the extreme forces and the fast movements in these sports; in climbing, however, heel hooks are often performed statically. Regarding the underlying pathologies of heel hook injuries around the knee, please refer to Sect. 11.5 (knee injuries). The selective use of imaging techniques is crucial in the supervision of climbing athletes. If an athlete reports a loud snapping noise described during a heel hook without corresponding persistent symptoms, imaging can be omitted with reference to the “snapping” of the iliotibial tractus. However, a clinical follow-up investigation is recommended. If the patient reports symptoms, an MRI of the knee joint and the distal thigh is recommended in addition to regular X-rays.

A sufficient warm-up before climbing/bouldering and regular stretching exercises are recommended to avoid heel hook injuries.

- **Etiology:** Instant pain during the heel hook movement, especially in bouldering. A painless loud snapping noise around the knee can occur during the heel hook.
- **Onset:** Acute onset of pain, usually on the dorsal thigh. Alternatively, snapping phenomenon around the knee, often painless.

- **Clinic:** Variable, depending on localization and underlying pathology.
- **Diagnosis:** Often by anamnesis (snapping phenomenon); in cases of complaints—ultrasound, X-ray, and MRI.
- **Therapy:** Depending on the underlying pathology.
- **Result:** Depends on pathology. Often complete healing, surgical therapy necessary occasionally.

11.5 Knee Injuries

While knee injuries are generally among the most common sports injuries, few injuries of this type have been diagnosed [9, 10].

However, the current worldwide indoor boulder “boom” is causing a rapid increase in the absolute number of injuries and especially acute injuries (see Chap. 2, Injury Statistics). Among these acute injuries, a recently growing number of knee injuries has been noticed. In this context, Josephsen et al. reported a higher prevalence of knee injuries in indoor bouldering compared to outdoor bouldering [11]. The increase in acute injuries around the knee is attributed to the design of modern climbing and bouldering facilities as well as the changed training methods in modern bouldering.

A comparative analysis of the current injury statistics (database of the authors’ patient collective) showed a doubling of knee injuries between 2015 and 2018 compared to the years 1998–2001. While 6.7% of the climbing-related injuries in the collective were in the knee joint area, other authors report percentages of up to 10% among the sport-specific injuries [12].

The underlying mechanisms of injury in bouldering and lead climbing have long been unclear and have recently been identified and published for the first time. The following four trauma mechanisms were identified: (1) high step



Fig. 11.8 High step position of the right lower extremity during bouldering with maximum flexion in the knee joint



Fig. 11.9 Drop knee position of the left lower extremity during bouldering with flexion in the knee joint and strong internal rotation in the hip joint

(Fig. 11.8), (2) drop knee (Fig. 11.9), (3) heel hook (Fig. 11.9), and (4) falls on the ground (Fig. 11.10) [13].

Meniscal lesions, strains of the iliotibial tract, and cruciate ligament injuries are the most common diagnoses. Individual mechanisms of injury and the resulting injuries were reported as follows (Table 11.1):



Fig. 11.10 Uncontrolled fall on the mat in bouldering

Table 11.1 Distribution of knee injuries [13]

Injury type	% in own patients
Medial meniscal tear	28.6
Iliotibial band sprain*	19.5
ACL tear +	9.1
LCL tear	7.8
Lateral meniscal tear	5.2
Patellar dislocation	3.9
Minor joint effusion	3.9
Cartilage injury	3.9
Others	3.9
ACL rupture	2.6
Partial ACL tear	2.6
Medial meniscal tear w. MCL tear	2.6
MCL tear	2.6
Popliteal muscle tear	2.6
Joint fracture	1.3

- *High step*: Mainly medial meniscus lesions
- *Drop knee*: Mainly medial/lateral meniscus lesions and medial collateral ligament injuries (Fig. 11.11)
- *Heel hook*: Mainly strains of the iliotibial tractus, the ischiocrural muscles, and lateral meniscus lesions (Fig. 11.12)
- *Falls on the ground*: Mainly ACL injuries

In terms of diagnostics and therapy, the supervision of climbing athletes does not differ significantly from that of other athletes. Results from the authors' patients showed an overall fast *return to sport* in climbing athletes after a knee injury.



Fig. 11.11 MRI of a lateral meniscus posterior horn lesion after a heel hook injury in bouldering (see arrow)

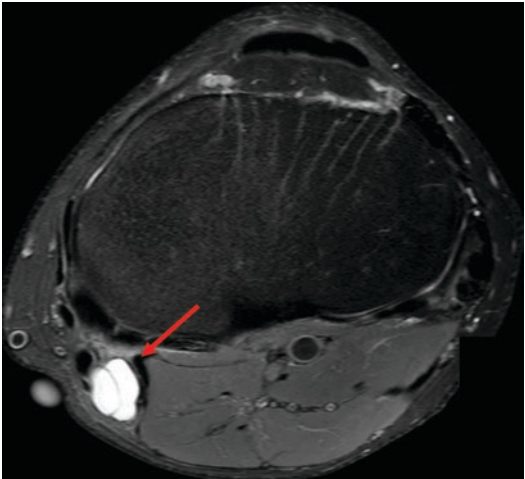


Fig. 11.12 MRI of a ganglion cyst at the ischiocrural muscles near the knee (see arrow)

Sport-specific stabilization exercises are recommended to prevent injuries from falls.

- **Etiology:** Four different trauma mechanisms: *high step, drop knee, heel hook, and fall on the ground*
- **Onset:** Acute injuries
- **Clinic:** Depending on injury
- **Diagnosis:** Clinical investigation, X-rays, and MRI
- **Therapy:** Often conservative therapy, surgical procedure required in approximately one-third of all injuries
- **Result:** Depending on injury; mostly good outcome

11.6 Conclusion

In summary, complaints in the hip/groin, thigh, and knee form a large group of clinical patterns in climbers. Sufficient knowledge of the underlying pathologies as well as mechanisms of injuries allows a quick and correct diagnosis. A conservative therapeutic approach is possible in many cases; acute injuries of the knee or specific anatomical variation around the hip sometimes require surgical therapy. Healing processes are sometimes prolonged, and adjustment of training after returning to sport is recommended.

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